

PHENIKAA UNIVERSITY
PHENIKAA SCHOOL OF COMPUTING



FINAL PROJECT REPORT
COURSE: SOFTWARE ARCHITECTURE
TASK MANAGEMENT SYSTEM

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CHAPTER I: PROJECT OVERVIEW

1.1 Problem Statement

In the modern digital landscape, the daily workflows of students, software engineers, and knowledge workers have become increasingly complex and heavily reliant on web-based applications. A typical study or work session requires users to interact with a multitude of digital resources simultaneously—ranging from learning management systems and documentation platforms (e.g., GitHub, StackOverflow) to design tools and project tracking dashboards. However, this heavy reliance on cloud-based tools has led to severe digital fragmentation. When transitioning from one assignment or project to another, individuals are forced to manually open, search for, and arrange numerous browser tabs.

This manual setup process is not merely an inconvenience; it imposes a significant cognitive penalty. Research in Human-Computer Interaction and cognitive psychology indicates that excessive application switching disrupts deep work and depletes mental energy before the actual core task even begins. Furthermore, examining the current software market reveals a noticeable limitation in existing productivity tools. Popular task management systems, such as Todoist, Trello, or Notion, operate purely as passive information repositories. They excel at reminding users what needs to be done (e.g., "Implement the UI storyboard for Project X"), but they remain entirely disconnected from the actual execution environment. They cannot proactively prepare the workspace required to perform that specific task.

To address this gap, the OmniDash project is proposed. OmniDash introduces the concept of "Workspace Combos"—a centralized system where a user's task is tightly coupled with its required web resources (URLs). By integrating a modern Web Application with a custom Browser Extension, OmniDash automates the environment setup. With a single click, the system securely transmits a payload of URLs to the browser extension,

which instantly launches the necessary tabs, thereby eliminating manual navigation and preserving user focus.

Developing such a distributed interaction model requires a robust underlying structure. A good software system must be architected correctly from the start, because the cost of fixing defects grows exponentially through development phases (Bass et al., 2021): a defect discovered during operation can cost up to 100 times more to fix than one discovered during design. Therefore, the primary research objective of this project is to design and implement OmniDash using a highly cohesive Layered Monolith Architecture. This ensures that the complex business logic of resource management and the secure event-driven communication between the web client and the browser extension are scalable, secure, and maintainable.

1.2 Project Objectives

The project aims to achieve the following concrete, measurable objectives based on SMART criteria:

1. Build a web application and browser extension that cover at least 4 core business functions: User Authentication, Task Management, Workspace Combo Management, and Automated Browser Tab Execution.
2. Apply the Layered Monolith Architecture as the primary architectural style, strictly following its principles of separation of concerns (Presentation, Business Logic, Persistence) and documenting all technical trade-offs using Architecture Decision Records (ADR).
3. Establish a secure and latency-optimized Message Passing mechanism between the web application and the Manifest V3 extension, ensuring a system response time of under 1.5 seconds for workspace activation.
4. Achieve at least 70% automated test coverage across the backend API endpoints and data access layers.

5. Complete the system development, integration testing, and a successful live demo within the 10-week academic schedule.

1.3 Project Scope

To ensure the project remains feasible within the given timeframe, the system boundaries are strictly defined as follows:

- In scope:
 - Secure user authentication and session management using JSON Web Tokens (JWT) and bcrypt password hashing.
 - A comprehensive Task Management module allowing users to perform full CRUD (Create, Read, Update, Delete) operations.
 - A Workspace Management module allowing users to group multiple URL resources under distinct functional categories.
 - A custom Google Chrome Browser Extension (Manifest V3) acting as a background agent to handle URL payloads and launch browser tabs automatically.
 - Seamless, secure one-way message passing from the web application client to the browser extension.
- Out of scope:
 - Native mobile applications (iOS/Android), as the core value of the system relies exclusively on desktop browser automation.
 - Real-time team collaboration, chat functionality, or multi-user workspace sharing.
 - Artificial Intelligence (AI) integrations for automatic task decomposition or contextual recommendations.
 - Integration with third-party APIs such as Google Calendar, GitHub, or payment gateways.

1.4 Technology Stack Overview

Layer	Technology	Rationale
Frontend	Next.js (React 18) + Zustand + Tailwind CSS	Component-based UI, efficient global state management, and rapid utility-first styling.
Backend	Node.js + Express.js	Lightweight, highly compatible with REST API design, and supports the Layered Monolith approach.
Database	PostgreSQL	Ensures ACID compliance and strong relational data integrity for tasks and resources.
ORM	Prisma	Provides type-safe database queries and built-in schema migration support.
Browser Extension	Manifest V3 API	The modern, secure standard mandated by Google for interacting with browser-level functionalities (chrome.tabs).
Diagramming & Design	Draw.io, Figma	Industry-standard tools for creating UML, C4 Models, and UI/UX storyboards.

CHAPTER II: REQUIREMENTS ANALYSIS

Chapter 2 constitutes the Software Requirements Specification (SRS)—the critical foundation for all subsequent design and architectural decisions. It rigorously defines what the OmniDash system must do (Functional Requirements) and how well it must perform under specific constraints (Non-Functional Requirements). All functional requirements are traceable to detailed use cases, while non-functional requirements are explicitly bound to measurable Quality Attribute Scenarios to ensure the system's architecture is driven by objective metrics.

2.1 Functional Requirements

The core functionalities of the OmniDash system are designed to solve the problem of digital fragmentation by linking task management with browser automation.

ID	Feature	Description	Priority
FR01	Login / Registration	The system allows users to register an account using an email and password. Passwords must be securely hashed. Authentication is maintained via JWT tokens.	High
FR02	Task Management	Authenticated users can perform full CRUD (Create, Read, Update, Delete) operations on tasks. Users can mark tasks as completed, set deadlines, and optionally link a task to a specific Workspace.	High
FR03	Workspace Combo Management	Users can create custom "Workspaces." Each Workspace serves as a container where users can add, edit, or remove multiple web resource URLs (e.g., GitHub, Google Docs, StackOverflow).	High
FR04	Workspace Activation (Automation)	Upon clicking "Activate" on a specific Workspace, the Web Client securely transmits the associated payload of URLs to the Browser	High

		Extension. The Extension then automatically opens all URLs in new browser tabs.	
FR05	Search & Filter	The system supports filtering tasks by their status (Pending/Completed) and searching for specific Workspaces by name, enabling quick navigation within the dashboard.	Medium

2.2 Non-Functional Requirements (NFR)

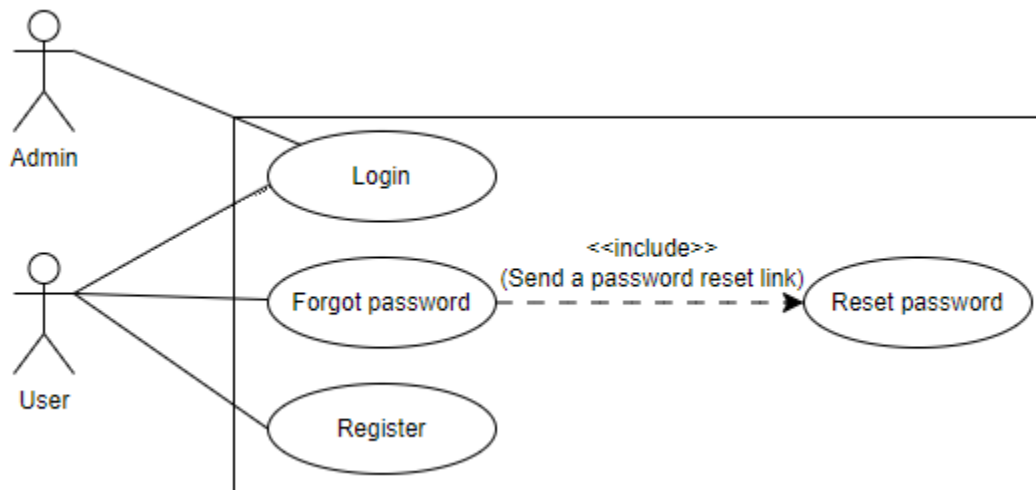
Non-functional requirements dictate the system's operational constraints. In OmniDash, these constraints act as architectural drivers.

Attribute	Requirement	Response Measure
Performance	Rapid Workspace Activation	The end-to-end response time (from Web UI click to the Browser Extension initiating tab creation) must be < 1.5 seconds for 95% of standard requests.
Availability	System Uptime	The backend REST API and Database must maintain an uptime of $\geq 99.5\%$ during business hours.
Security	Authentication & Data Protection	JWT Bearer tokens must be used for API endpoints. Passwords must be hashed using bcrypt with salt rounds ≥ 10 . No cross-origin payload execution.
Integrability	Web-Extension Messaging	The system must guarantee reliable payload delivery from the Next.js Web App to the Manifest V3 Extension without triggering browser popup blockers.
Maintainability	Code Quality & Layering	Strict separation of concerns (Layered Monolith); test coverage $\geq 70\%$; utilization of Prisma ORM for safe database migrations.

Usability	Minimalist UI Responsiveness	The Next.js web interface must render seamlessly on standard laptop/desktop monitors (1024px to 1920px), prioritizing focus.
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2.3 Use Case Diagram

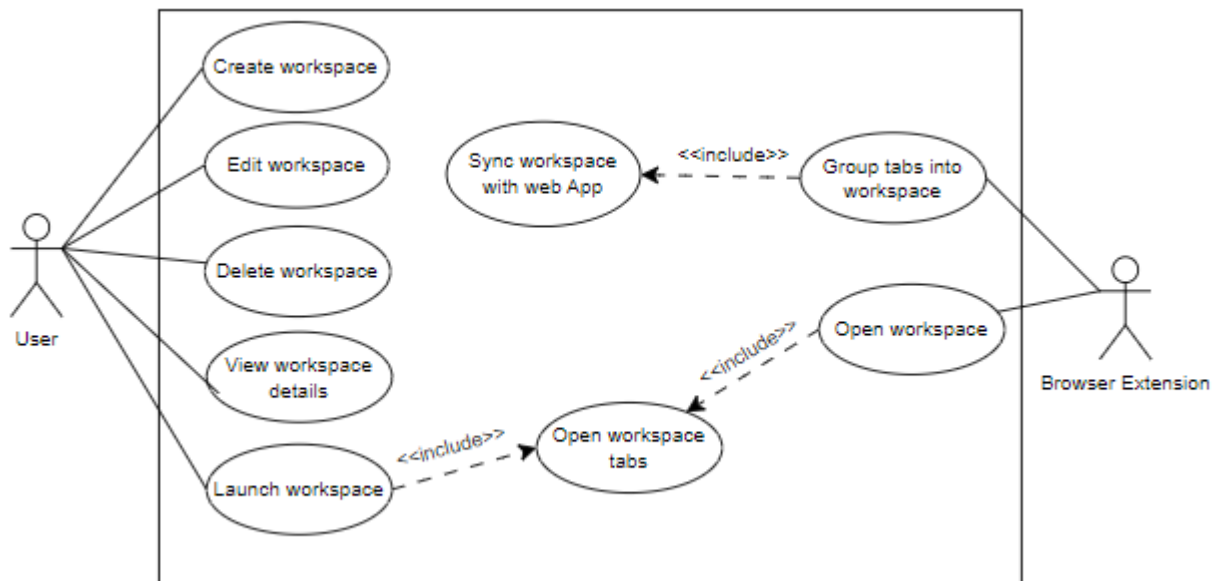
- Use Case: User Authentication



Field	Content
Use Case ID	UC-01
Use Case Name	User Authentication (Login)
Primary Actor	Admin, User
Preconditions	The actor possesses a valid registered account and has an active internet connection.
Postconditions	The system authenticates the user, issues a JWT session token, and redirects them to the main Dashboard.
Main Flow	1. Actor navigates to the Login page.

	2. Actor inputs their Email and Password. 3. System securely validates the credentials against the database. 4. System grants access and redirects the actor based on their Role (Admin/User).
Exception Flow	3a. Incorrect password: The system denies access and displays an "Invalid Credentials" error message. 3b. Forgotten password: The actor selects "Forgot Password" -> The system initiates the recovery email sequence (triggers <<include>> Reset Password).

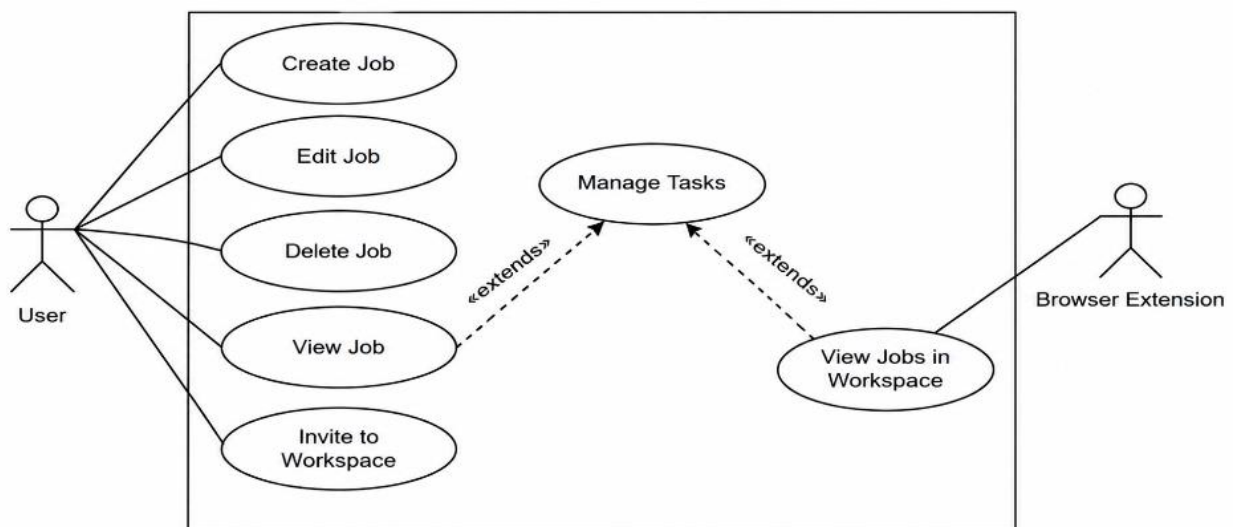
- Use Case: Workspace



Field	Content
Use Case ID	UC-02
Use Case Name	Activate Workspace

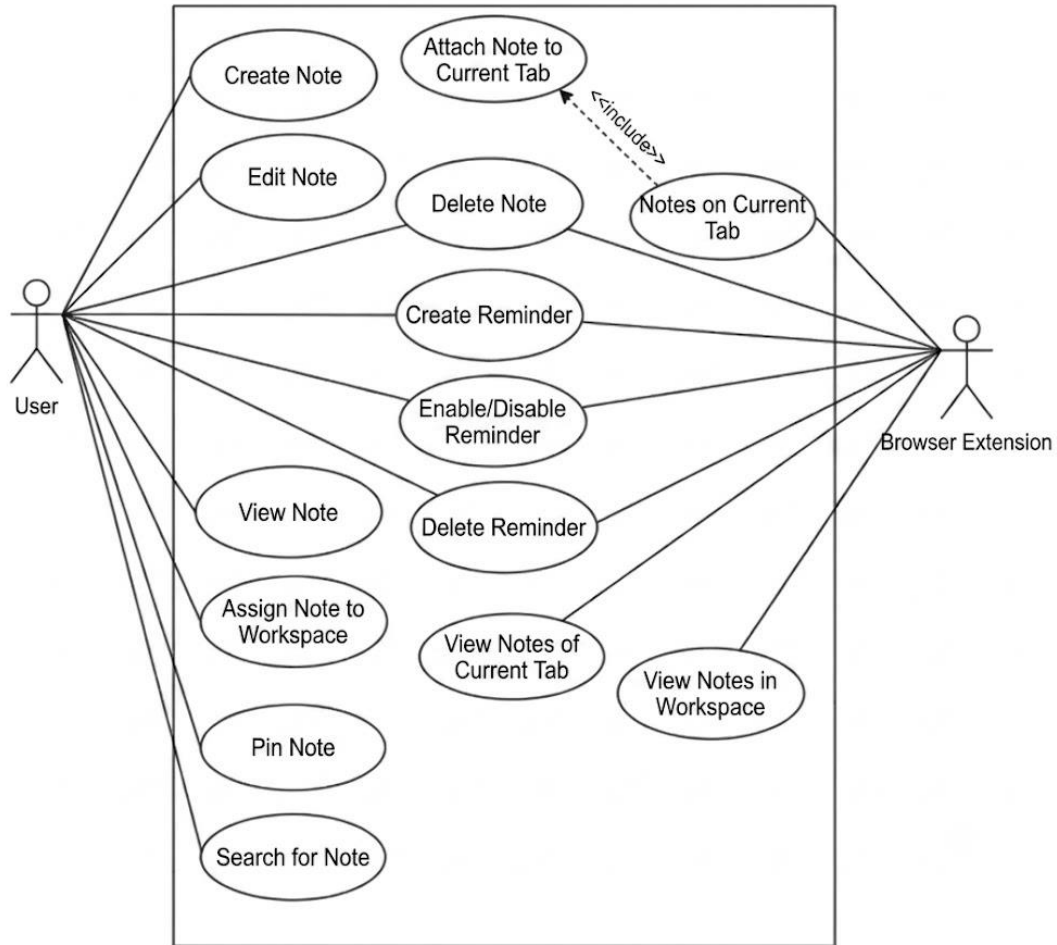
Primary Actor	Logged-in User, Browser Extension
Preconditions	The user is authenticated on the Web App; the target Workspace contains at least one URL; the Browser Extension is installed and enabled.
Postconditions	All URLs associated with the selected Workspace are successfully launched in new browser tabs.
Main Flow	1. User clicks the "Activate Workspace" button on the Web UI. 2. Web App packages the URL list into a payload and emits a message to the Browser Extension. 3. Browser Extension intercepts the message and executes the <<include>> Open Workspace Tabs action (invoking the chrome.tabs.create API). 4. Web App displays a success notification to the user.
Exception Flow	2a. Extension missing or disabled: The Web App receives no acknowledgment, times out, and displays an error prompting the user to install or enable the extension.

- Use Case: Manage job



Field	Content
Use Case ID	UC-03
Use Case Name	Manage Job
Primary Actor	User
Preconditions	The user is successfully authenticated and has accessed the Job Management interface.
Postconditions	Jobs are successfully viewed, created, updated, or deleted, with data synchronized in the PostgreSQL database.
Main Flow	1. User accesses the Job Management dashboard. 2. System retrieves and displays the active job list. 3. User can choose to add a new job, modify an existing job's details, or toggle its completion status. 4. System validates the operations and updates the database records via Prisma ORM. 5. The interface dynamically re-renders to reflect the updated job states.
Alternative Flow	3a. User chooses to delete a job: The system prompts for confirmation, executes the deletion query, and removes the item from the UI list.
Exception Flow	4a. Network disconnection occurs during data submission: The system displays an error toast stating "Sync failed" and keeps the local changes pending.

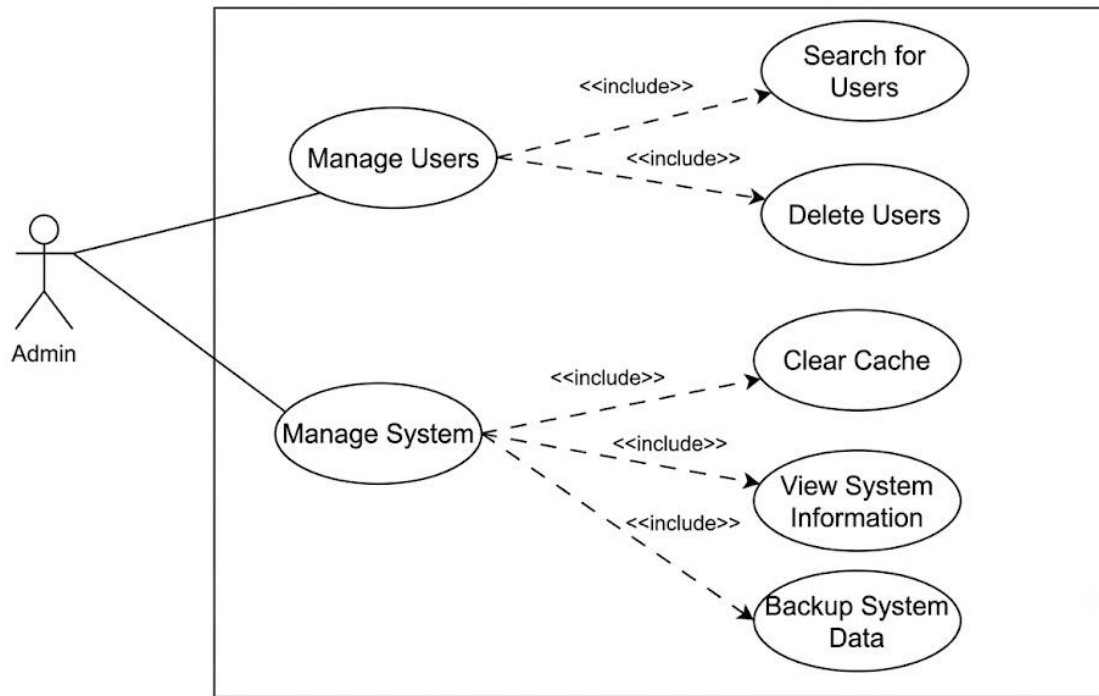
- Use Case: Note and Reminder



Field	Content
Use Case ID	UC-04
Use Case Name	Manage Note and Reminder
Primary Actor	User, Browser Extension
Preconditions	The user possesses a valid active session token and enters the Note & Reminder workspace.
Postconditions	Notes and their associated time-based reminders are properly configured, modified, or cleared from the system.

Main Flow	1. User views the list of existing notes and active reminders on the workspace dashboard. 2. User can write a new note, edit existing text contents, or pin/unpin notes for better layout organization. 3. User can select a specific note to attach, enable, or disable a time-based notification reminder. 4. System saves the combined data payload (note content and reminder configuration) to the backend storage. 5. The dashboard and browser extension popup refresh to display the updated statuses.
Alternative Flow	3a. User opens the browser extension popup on an active webpage: The system filters and displays notes and reminders specific to that active URL tab context.
Exception Flow	3b. User sets a reminder time that has already passed: The system blocks the activation and highlights the date-time picker with a validation error.

- Use Case: Manage users and system



Field	Content
Use Case ID	UC-05
Use Case Name	Manage users and system
Primary Actor	Admin
Preconditions	The logged-in account possesses Administrator privileges (Role = ADMIN).
Postconditions	User records in the system are successfully retrieved, filtered, or securely removed from the database.
Main Flow	1. Admin accesses the User Management dashboard. 2. System fetches and displays a paginated list of all registered users.

	3. Admin enters a keyword in the search bar, triggering the <<include>> Search User action. 4. Admin selects a violating account and clicks "Delete", triggering the <<include>> Delete User action. 5. System prompts for final confirmation, executes the deletion query, and reloads the active list.
Exception Flow	4a. Attempting to delete another Administrator account: The backend rejects the request, and the system displays an unauthorized permission warning (HTTP 403 Forbidden).

2.4 Architecturally Significant Requirements (ASRs)

2.4.1 Identifying ASRs: The Utility Tree

The standard method for eliciting and prioritizing ASRs is the Utility Tree. The tree decomposes the system's overall value into quality attributes, then into specific attribute refinements, and finally into concrete, testable scenarios.

Quality Attr.	Refinement	ASR Scenario	Biz Val.	Arch. Imp.
Integrability	Cross-Environment Communication	The Web Application must reliably send a JSON payload of URLs to the Browser Extension without triggering cross-origin or pop-up blocker security mechanisms.	H	H
Performance	Rapid Automation	Activating a Workspace with 10 URLs must invoke the browser tab creation process within < 1.5 seconds at the 95th percentile.	H	M
Modifiability	Layer Isolation	A developer must be able to modify the database schema (e.g., adding a new field to Tasks)	M	H

		without altering the API Controllers or Frontend logic, completed in ≤ 1 day.		
Security	API Authorization	An unauthenticated entity cannot access any endpoint. All backend requests must be validated via JWT Bearer tokens with strict ownership checks.	H	H
Availability	Error Recovery	If the PostgreSQL database connection drops, the Node.js backend must not crash but instead return graceful HTTP 500 errors until the connection is restored.	M	M
Usability	UI Responsiveness	The Dashboard UI must render correctly and maintain functional layout integrity on all desktop viewports from 1024px to 1920px.	H	L

2.4.2 ASR Table (Mapping to Architecture Drivers)

After identifying ASRs via the Utility Tree, they are mapped to concrete architecture drivers to justify the structural decisions made in Chapter 3.

ID	ASR Scenario (Summary)	Quality Attr.	Architecture Driver
ASR01	Reliable Web-to-Extension payload transfer (H,H)	Integrability	Event-Driven Message Passing via window.postMessage combined with Manifest V3 background.js listeners.
ASR02	Unauthenticated users blocked; JWT validation (H,H)	Security	Stateless REST API with custom Express.js Authentication Middleware & bcrypt hashing.

ASR03	Schema modifications isolated from business logic (M,H)	Modifiability	Implementation of Layered Monolith Architecture utilizing Prisma ORM as the abstract Persistence Layer.
ASR04	< 1.5s execution for 10 URLs (H,M)	Performance	Asynchronous JavaScript execution (Non-blocking I/O in Node.js) and lightweight Zustand state management on the Frontend.

CHAPTER III: ARCHITECTURE & SYSTEM DESIGN

3.1 Architectural Style Selection

3.1.1 Theoretical Foundation

Software architecture is fundamentally organized around two core concepts:

- **Component:** An independently deployable unit with a well-defined interface, encapsulating a subset of system functionality.
- **Connector:** The communication mechanism facilitating interaction between components (e.g., REST API calls, procedure calls, or message passing).

To ensure high maintainability, the system's design distinguishes between high-level Architectural Styles (the overarching system organization) and Architectural Patterns (concrete, localized solutions like Dependency Injection or the MVC pattern).

3.1.2 Common Architectural Styles and Selection Rationale

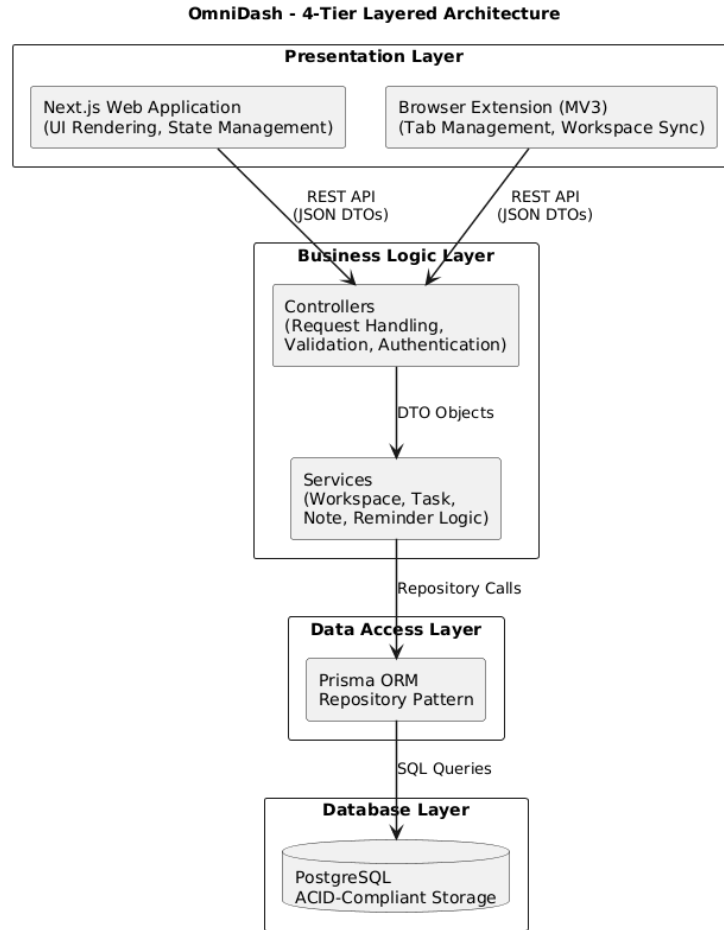
While evaluating architectural styles for OmniDash, Microservices and Layered Architecture were considered. Microservices offer independent scaling but introduce significant operational overhead (DevOps, network latency, distributed transactions). Given that OmniDash centers around a highly cohesive domain (Task and Workspace management) and is developed by a small team within a 10-week timeframe, a distributed system would be an over-engineering anti-pattern.

Therefore, the Layered Monolith Architecture was selected as the primary architectural style. It organizes the system into functional layers where upper layers depend strictly on lower layers. This ensures a strong separation of concerns, simplifies the deployment pipeline, and makes the codebase highly testable and modifiable.

3.2 Overall System Architecture

The OmniDash system adopts a strict 4-Tier Layered Architecture. Data is passed between these layers exclusively using Data Transfer Objects (DTOs), and layer-skipping (e.g., the Presentation layer querying the database directly) is strictly prohibited to maintain system integrity.

- **Presentation Layer (Frontend & Extension):** Built with Next.js (React 18) and a Manifest V3 Browser Extension. This layer is responsible for rendering the minimalist UI, managing global client states via Zustand, capturing user interactions, and securely transmitting payload messages (`window.postMessage`) to the extension. Rule: This layer contains zero core business logic.
- **Business Logic Layer (Backend Controller & Service):** Built with Node.js and Express.js. It acts as the brain of the system. The Controller sub-layer parses HTTP requests and validates inputs, while the Service sub-layer executes complex business rules, such as aggregating URLs for Workspace Combos and enforcing JWT authorization.
- **Data Access Layer (Persistence):** Implemented using Prisma ORM. It encapsulates all complex SQL query logic and provides an abstract, type-safe Repository interface for the Business Logic Layer. Rule: Upper layers only know the repository interfaces, keeping the business logic entirely database-agnostic.
- **Database Layer:** Powered by PostgreSQL. It provides persistent, relational, and ACID-compliant data storage for Users, Tasks, Workspaces, and Resource URLs.



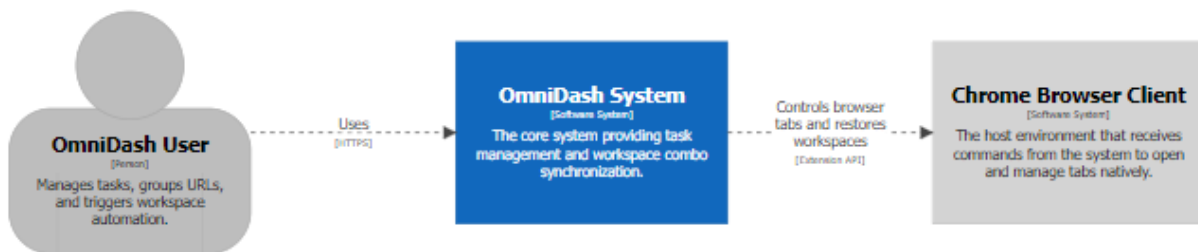
3.3 C4 Model – Architecture Documentation

The C4 Model (Simon Brown) is utilized to document the OmniDash software architecture. By providing distinct levels of abstraction, it guarantees that all stakeholders—from business reviewers to backend developers—can comprehend the system's structural decisions without being overwhelmed by unnecessary technical details.

3.3.1 C4 Level 1 – System Context Diagram

- **Purpose:** To illustrate the macroscopic view of the OmniDash ecosystem, identifying the system boundaries, its primary users, and its interactions with external environments.
- **Audience:** Business stakeholders, instructors, and product owners.

Actor / System Type	Name	Relationship Description
Person (User)	OmniDash User	Manages tasks, groups URLs, and triggers workspace automation.
Software System	OmniDash System	The core system providing task management and workspace combo synchronization.
External System	Chrome Browser Client	The host environment that receives commands from the system to open and manage tabs natively.

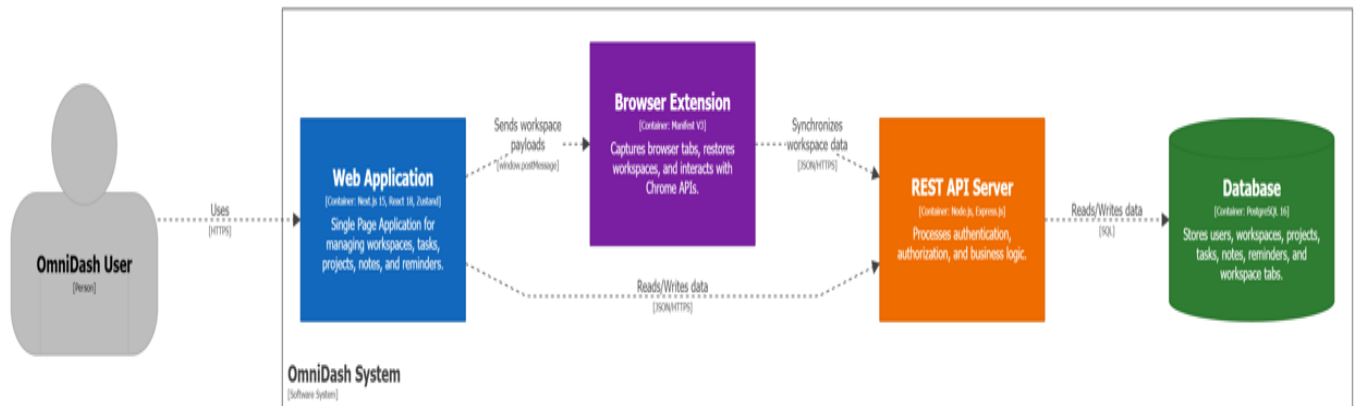


3.3.2 C4 Level 2 – Container Diagram

- **Purpose:** To zoom into the OmniDash System boundary, revealing the independently runnable/deployable containers and how they communicate.
- **Audience:** Software architects and DevOps engineers.

Container	Technology	Responsibility & Interaction
Web Application	Next.js, React 18	The SPA delivered to the user's browser. Makes JSON/HTTPS requests to the API and emits DOM messages to the Extension.
Browser Extension	Manifest V3 (background.js)	Runs natively in the browser. Listens for URL payloads from the Web App and triggers browser-level APIs (chrome.tabs).

REST API Server	Node.js, Express.js	The Monolithic backend processing authentication, business rules, and serving data to the SPA.
Database	PostgreSQL 16	The relational database storing all user, task, and workspace data.



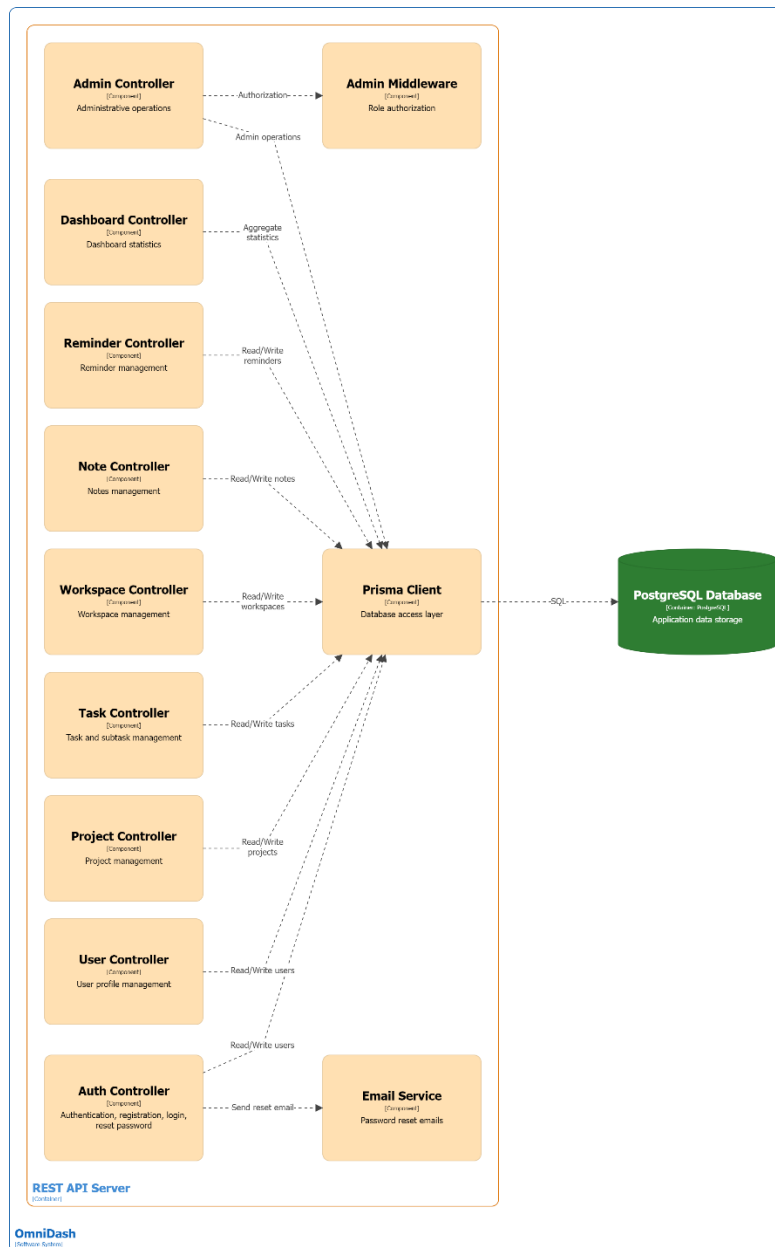
3.3.3 C4 Level 3 – Component Diagram (REST API Container)

- **Purpose:** To illustrate the internal structure of the REST API Server container and show how the Layered Monolith architecture is organized into controllers, middleware, infrastructure components, and database access layers.
- **Audience:** Backend developers, software architects, and maintainers responsible for implementing and extending the server-side application.

Component	Technology	Responsibility
AuthController	Express Controller	Handles authentication-related endpoints, including user registration, login, JWT-based authentication, password reset, and account recovery workflows.

UserController	Express Controller	Manages user profile retrieval, updates, and user-related operations.
ProjectController	Express Controller	Handles project creation, modification, deletion, and project organization features.
TaskController	Express Controller	Manages tasks, subtasks, priorities, statuses, hierarchical task structures, and task sequencing.
WorkspaceController	Express Controller	Handles workspace creation, configuration, membership management, and workspace-related operations.
NoteController	Express Controller	Provides CRUD operations for notes associated with users and workspaces.
ReminderController	Express Controller	Manages reminder scheduling, updates, notifications, and reminder-related business logic.
DashboardController	Express Controller	Aggregates project, task, and workspace data to provide dashboard statistics and summaries.
AdminController	Express Controller	Handles administrative operations and system-level management functions.
AdminMiddleware	Express Middleware	Performs role-based authorization and protects administrator-only endpoints.
Prisma Client	Prisma ORM	Acts as the data access layer, executing queries and transactions against PostgreSQL and

		abstracting database interactions.
Email Utility	Nodemailer Utility	Sends password reset emails and other system-generated notifications.
PostgreSQL Database	PostgreSQL	Stores persistent application data, including users, projects, tasks, workspaces, notes, reminders, and administrative information.



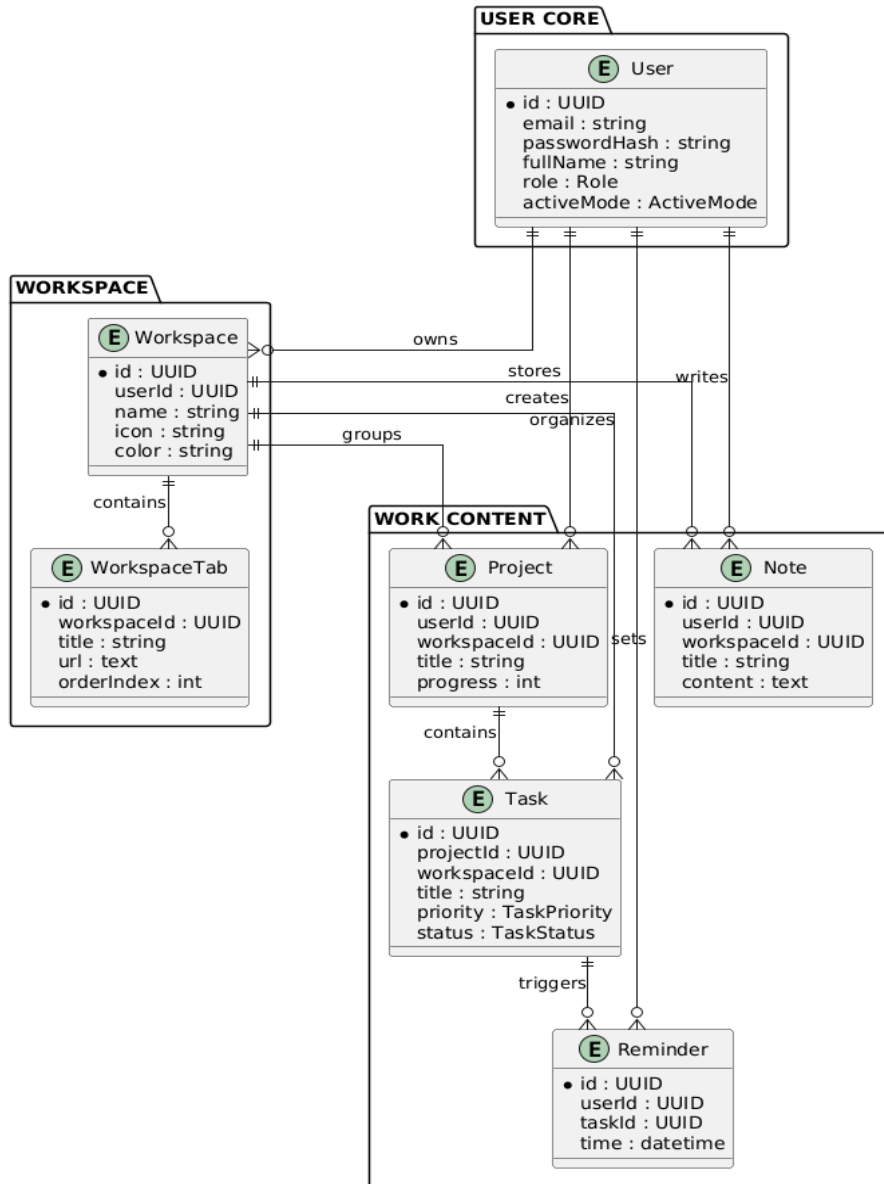
3.4 Database Design

The data persistence layer of OmniDash relies on a relational database model to ensure strict data integrity and enforce relationships between users, tasks, and their associated workspaces.

3.4.1 ERD (Entity-Relationship Diagram)

The database schema consists of four primary entities: User (1:N relationship with Tasks and Workspaces), Workspace (1:N relationship with Resources and Tasks), Task, and Resource (URL entities).

ERD - OmniDash (Optimized Layout)



3.4.2 Data Dictionary

- User

Field	Data Type	Description
Id	UUID	Primary key
email	String	Login email (unique)
passwordHash	String	Hashed password
fullName	String	Full name
role	Enum	User role
activeMode	Enum	Current active mode

- Workspace

Field	Data Type	Description
Id	UUID	Primary key
userId	UUID	User reference
Name	String	Workspace name
Icon	String	Workspace icon
Color	String	Display color

- **WorkspaceTab**

Field	Data Type	Description
Id	UUID	Primary key
workspaceId	UUID	Workspace reference
Title	String	String Tab name
url	Text	Website URL
orderIndex	Int	Tab display order

- **Project**

Field	Data Type	Description
Id	UUID	Primary key
userId	UUID	User reference
workspaceId	UUID	Workspace reference
Title	String	Project name
Progress	Int	Completion progress

- **Task**

Field	Data Type	Description
Id	UUID	Primary key
projectId	UUID	Project reference

workspaceId	UUID	Workspace reference
Title	String	Task name
Priority	Enum	Priority level
Status	Enum	Task status

- **Note**

Field	Data Type	Description
Id	UUID	Primary key
userId	UUID	User reference
workspaceId	UUID	Workspace reference
Title	String	Note title
Content	Text	Note content

- **Reminder**

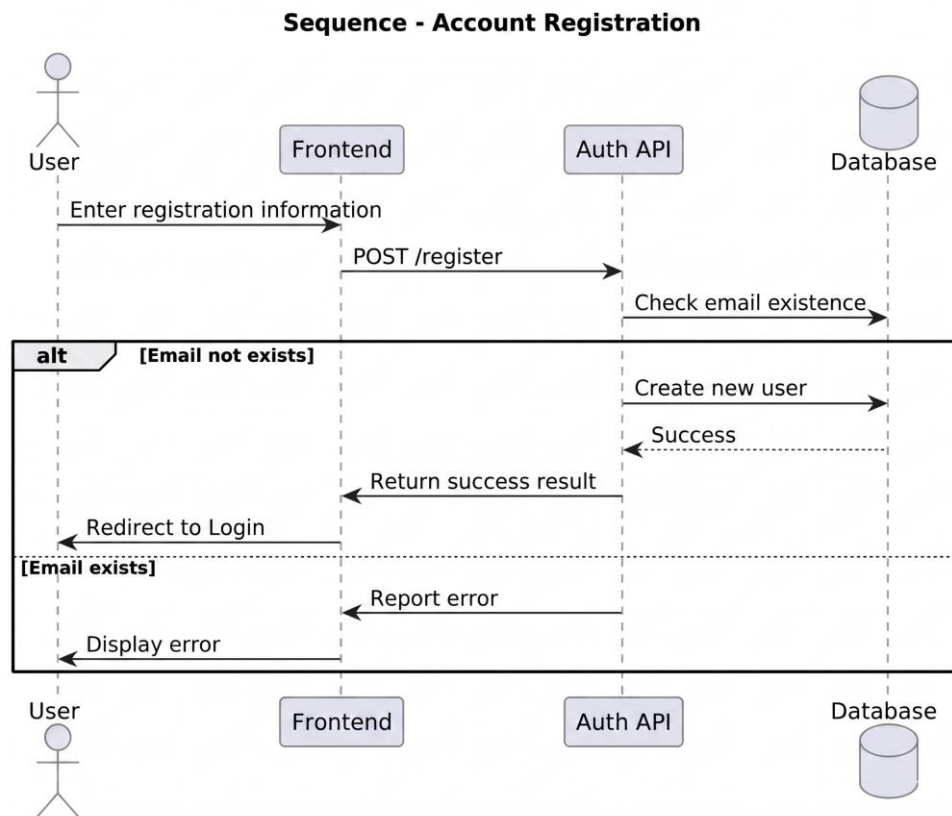
Field	Data Type	Description
Id	UUID	Primary key
userId	UUID	User reference
taskId	UUID	Task reference
Time	Datetime	Reminder time

3.5 UML Diagrams

3.5.1 Sequence Diagrams

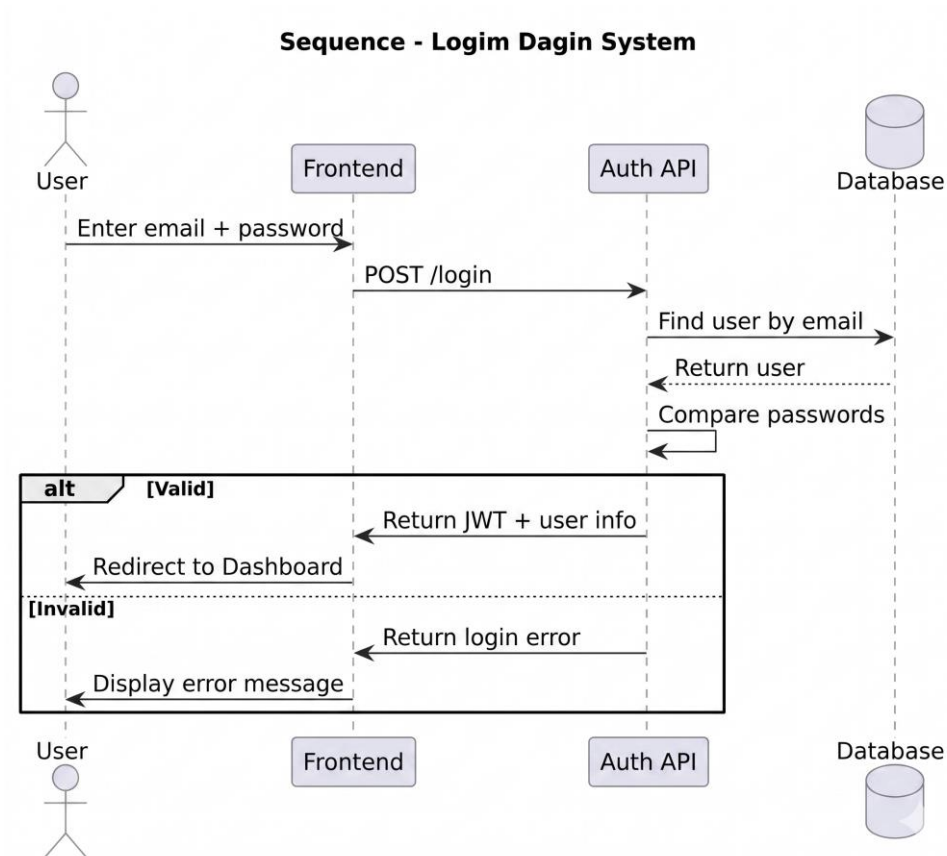
- **Register**

This sequence diagram illustrates the user registration flow. It shows the Frontend sending a registration request to the Auth API, which checks the database to see if the email already exists. It handles two scenarios: successfully creating a new user and redirecting to login, or returning an error if the email is already in use.



- **Login**

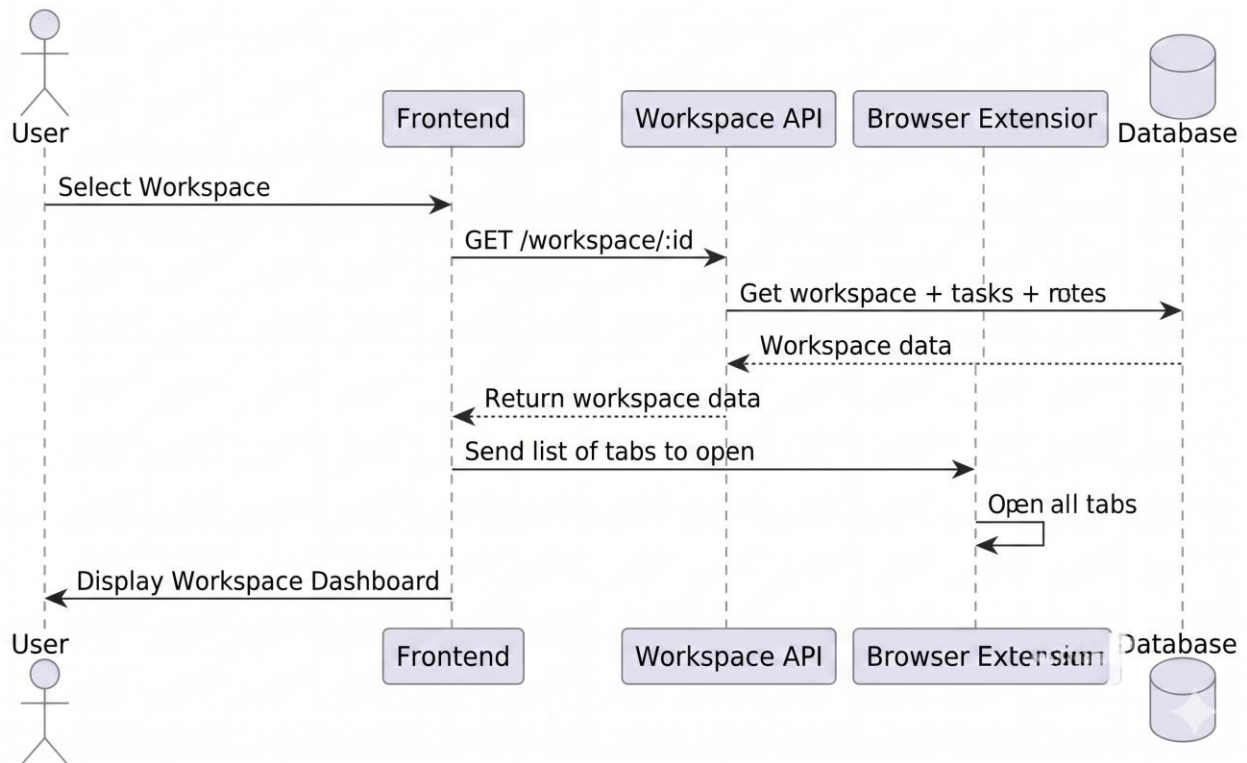
This diagram depicts the authentication process. The user submits their credentials, which the Auth API verifies against the database. If valid, the system returns a JWT (JSON Web Token) and redirects the user to the Dashboard. If invalid, it returns an error message to be displayed on the Frontend.



- **Open workspace**

This diagram shows the process of loading a selected workspace. The Workspace API fetches the workspace data, including tasks and notes, from the database. It then sends the list of saved URLs to the Browser Extension, which automatically opens all the associated tabs while the Frontend displays the Workspace Dashboard.

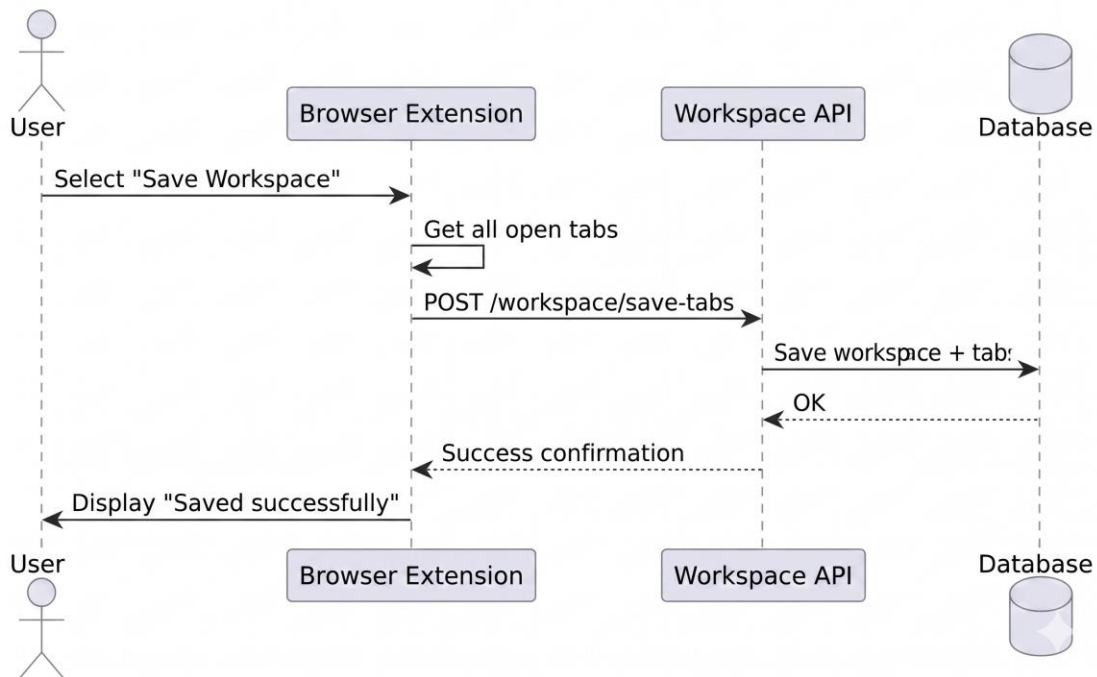
Sequence - Open Workspace



- **Save workspace from extension**

This sequence outlines how a user saves their current browsing session. When initiated by the user, the Browser Extension captures all currently open tabs and sends a request to the Workspace API, which then saves the workspace and tab data into the database and confirms the successful save back to the user.

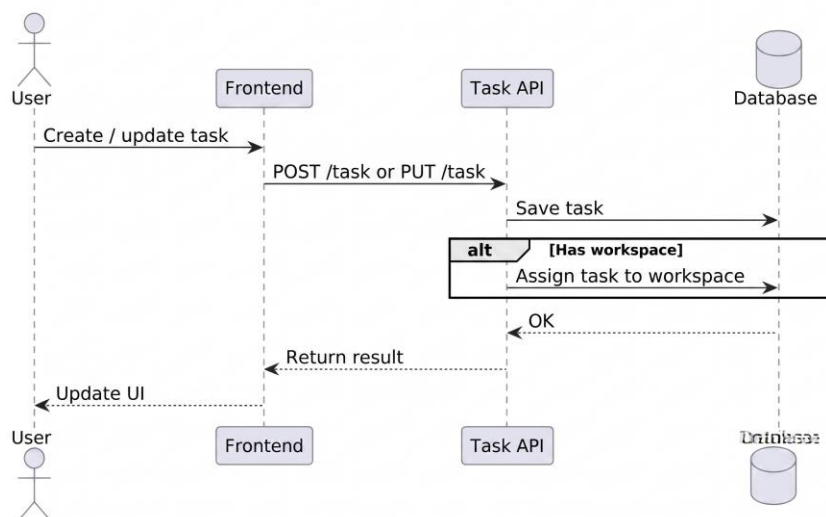
Sequence - Save Workspace from Extension



- **Task management**

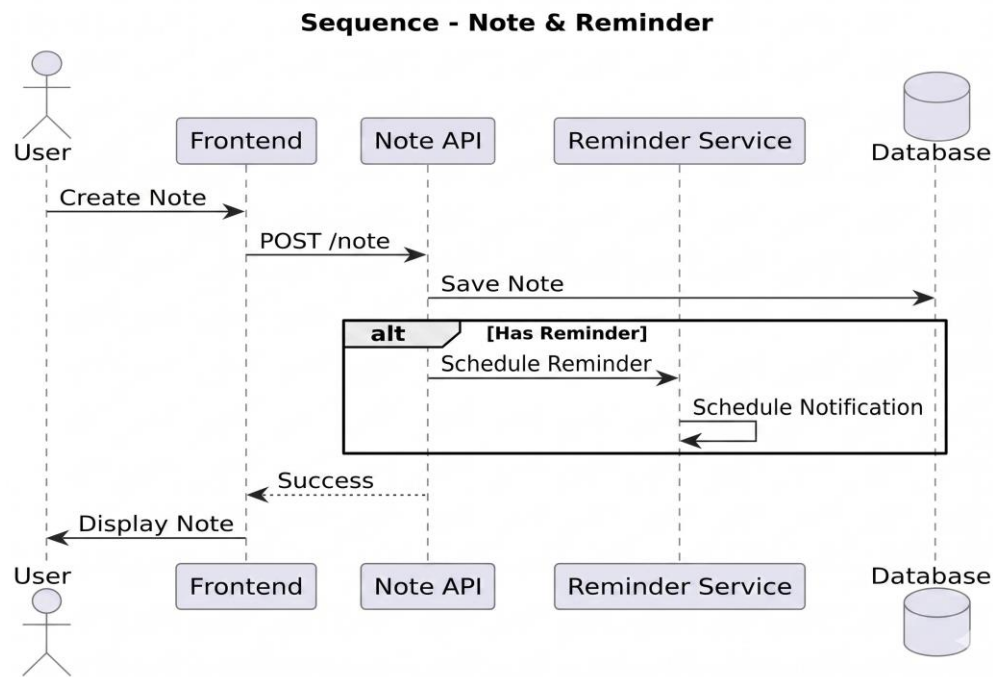
This diagram explains the flow for creating or updating a task. The Frontend sends the task details to the Task API to be saved in the database. It includes an optional block (alt) showing that if the task belongs to a specific workspace, it is assigned to that workspace before returning the success response to update the user interface.

Sequence - Task Management



- **Note and reminder**

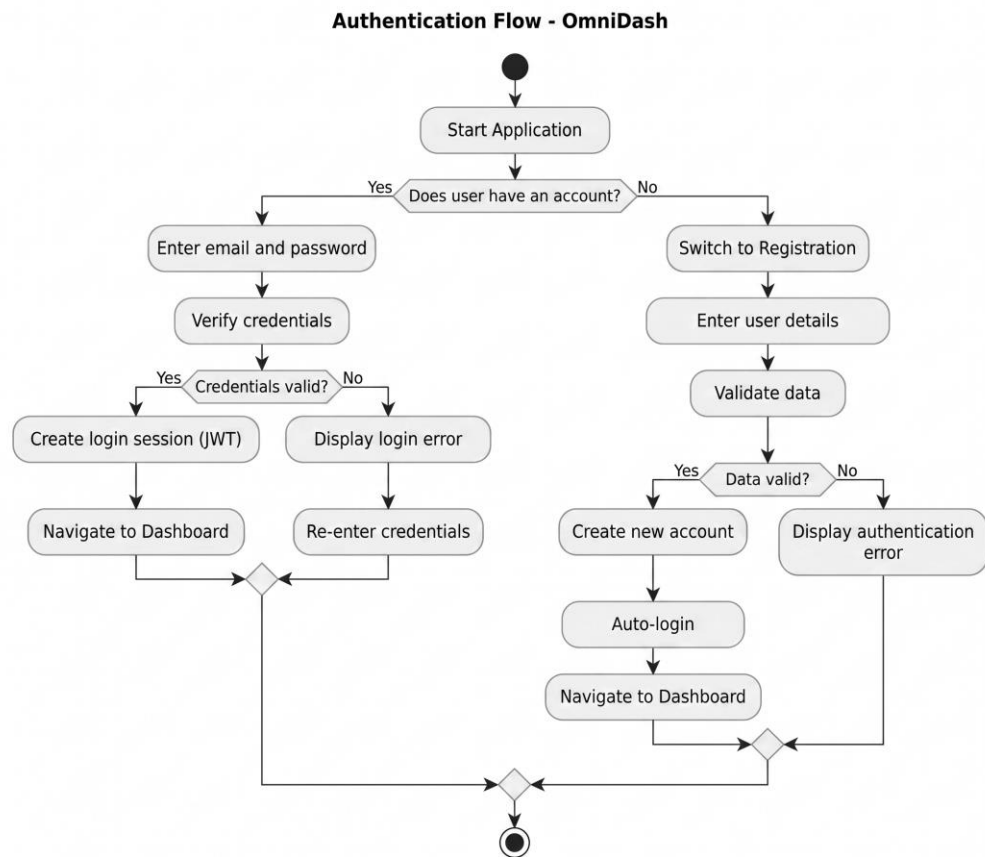
This diagram details the process of creating a note with an optional reminder. The Note API saves the note content to the database. If the user set a reminder time, it triggers a separate Reminder Service to schedule a notification before returning a success message to the Frontend.



3.5.2 Activity Diagrams

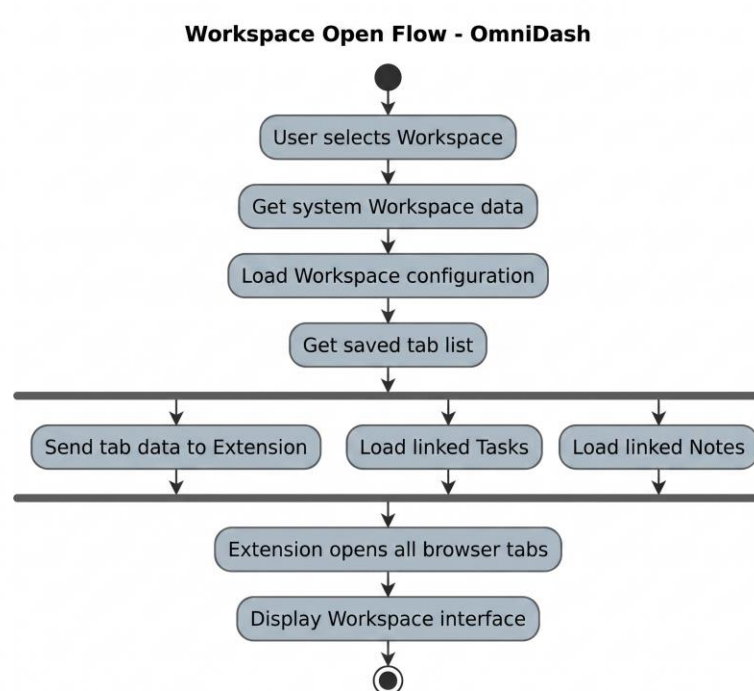
- **Authentication Flow**

This flowchart outlines the process for user authentication. It manages branching logic depending on whether the user is logging into an existing account (via credential verification and session creation) or registering a new account (via validation, creation, and auto-login), resulting in access to the Dashboard or displaying errors.



- **Workspace Open Flow**

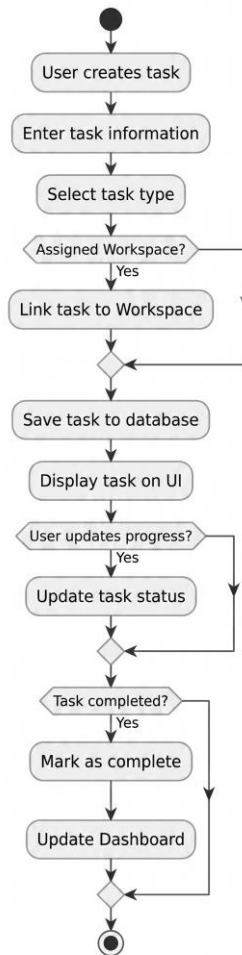
This diagram illustrates the sequence of actions initiated when a user selects a Workspace. It involves fetching configurations and data, followed by parallel processes to send tab data to the Extension, load linked tasks, and load linked notes. The flow concludes with the Extension opening browser tabs and displaying the complete Workspace interface.



- **Task Management Flow**

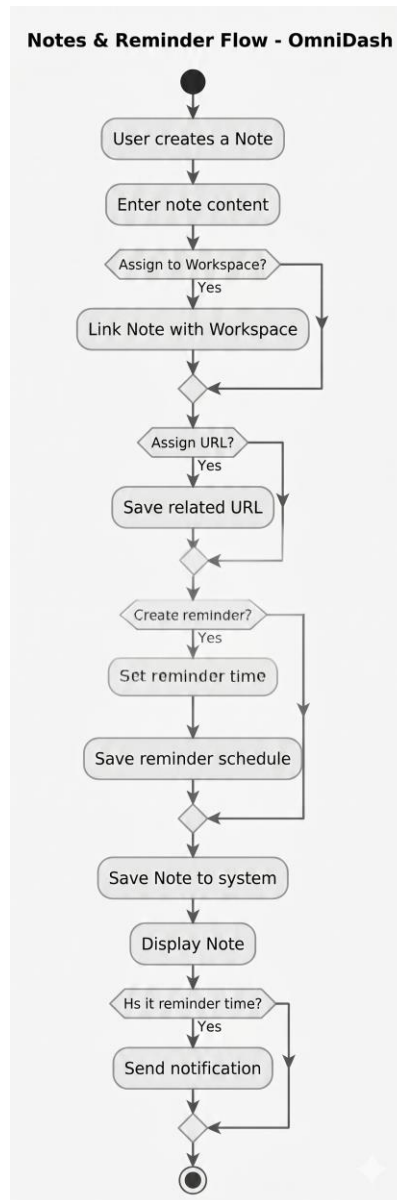
This flowchart defines the lifecycle operations for managing tasks. It traces the creation of a task with specific details, the decision logic for linking to a Workspace and saving data, followed by ongoing loops for updating task status based on progress, and the final process of marking it complete and updating the main Dashboard.

Task Management Flow - OmniDash



- **Notes & Reminder Flow**

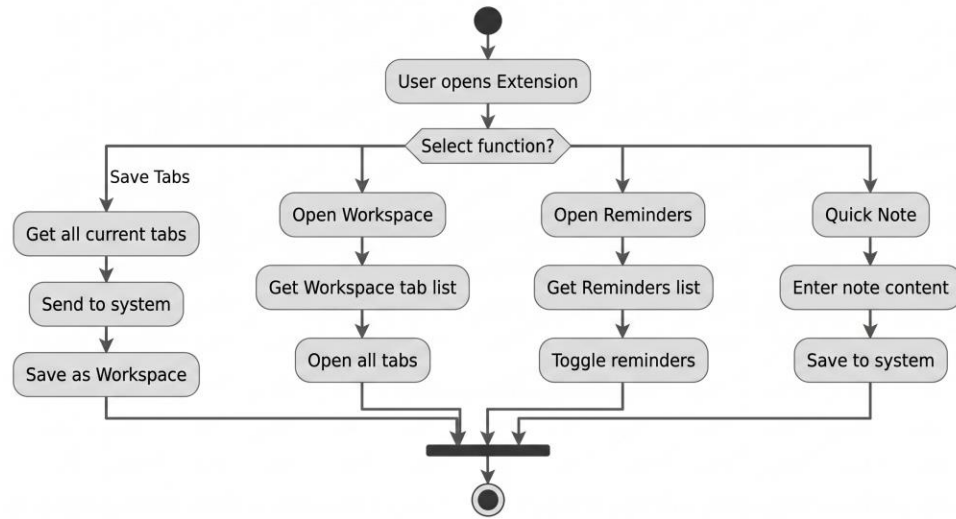
This diagram describes the complete process for creating and managing notes. It covers entering content, decision paths for linking the note to Workspaces or URLs, setting optional reminder times, saving to the system, displaying the note, and finally, sending a notification when the reminder time arrives.



- **Browser Extension Flow**

This diagram depicts the immediate functional choices available to a user after opening the browser extension. Based on the user's selection, the flow branches into one of four distinct operations: saving current tabs as a workspace, opening an existing workspace, managing reminders, or creating quick notes, with each process concluding after execution.

Browser Extension Flow - OmniDash



CHAPTER IV: IMPLEMENTATION & CODING

4.1 Development Environment & Tools

Category	Tool / Version	Purpose	Link
Backend Language	Node.js 20 LTS	JavaScript runtime for the REST API server	nodejs.org
Backend Framework	Express.js 4.x	Routing, middleware, and HTTP request handling	expressjs.com
Frontend Framework	Next.js 14 (React 18)	Component-based, optimized Web UI rendering	nextjs.org
Frontend Language	TypeScript 5.x	Type-safe static checking for the frontend	typescriptlang.org
Database	PostgreSQL 16	Relational data storage for core entities	postgresql.org
ORM	Prisma 5.x	Type-safe database queries and migrations	prisma.io

Browser API	Manifest V3	Standard for developing the Chrome Extension	developer.chrome.com
Version Control	Git / GitHub	Source code management and version tracking	github.com
Diagram Tool	Draw.io	Creation of C4 Models, UML, and ERD	diagrams.net

4.2 Project Directory Structure

The repository is structured as a monorepo containing three distinct execution environments: the Backend, the Frontend, and the Browser Extension. The backend directory strictly enforces the Layered Architecture.

```

+---backend
|   \---prisma
|       \---migrations
|       \---src
|           +---config
|           +---controllers
|           +---middlewares
|           +---routes
|           \---utils
+---Document
+---extension
\---frontend
    \---app
        +---admin
        |   +---system
        |   |   \---users
        |   +---components
        |   +---dashboard
        |       +---calendar
        |       +---notes
        |       +---project
        |       |   \---[id]
        |       +---projects
        |       +---reminders
        |       +---settings
        |       \---workspaces
        +---forgot-password
        +---login
        +---register
        +---reset-password
        |   \---[token]
        \---store

```

4.3 Notable Implementation Techniques

- **Technique 1: Secure JWT Authentication Pipeline**

- **Description:** OmniDash employs a stateless authentication mechanism. Upon successful login, the backend generates a JSON Web Token (JWT). To protect subsequent API calls, an Express middleware intercepts requests, extracts the Bearer token, and validates the signature.
- **Architectural Implementation:** Instead of embedding authentication checks within individual endpoints, the system utilizes a centralized Express.js Middleware. This middleware intercepts incoming requests before they reach the Controller layer. It extracts the token, verifies the cryptographic signature using a server-side secret key, and attaches the decoded user payload to the request object. If the token is missing, expired, or tampered with, the middleware immediately rejects the request with an HTTP 401 Unauthorized status.
- **Analysis:** This implementation enforces the Single Responsibility Principle (SRP). By extracting the authorization logic into a dedicated middleware layer, the Presentation Layer (Controllers) remains entirely focused on routing and parsing Data Transfer Objects (DTOs), preventing code duplication and shielding the core business logic from unauthenticated intrusions.

- **Technique 2: Cross-Environment Message Passing (Web to Extension)**

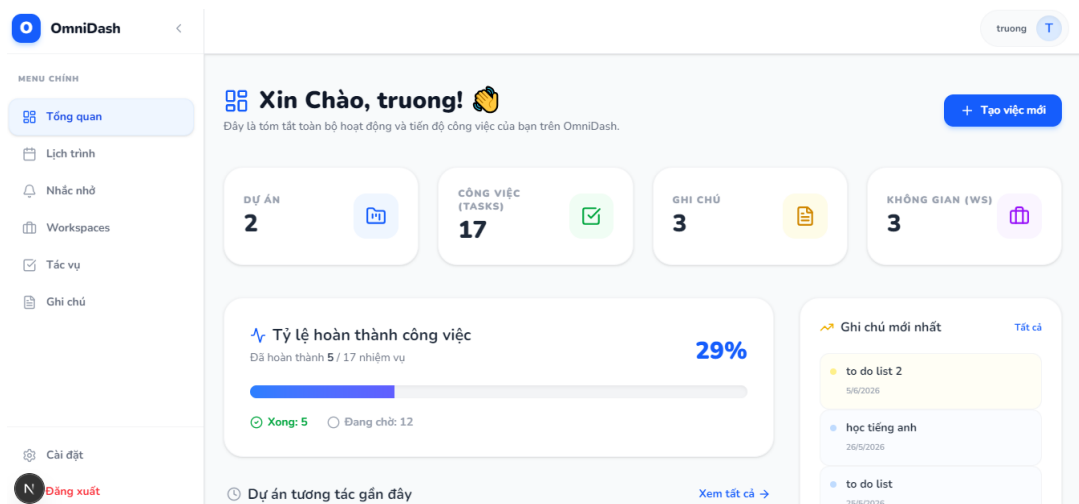
- **Description:** To satisfy the primary Architecturally Significant Requirement (ASR-01), the Next.js Web App must trigger browser tabs without being blocked by pop-up blockers. This is achieved via Event-Driven message passing.
- **Architectural Implementation:** The system bridges the Next.js Web App and the Manifest V3 Extension using an Event-Driven message-passing pattern. When a user clicks "Activate Workspace", the web client fetches the URL list from the backend and packages it into a structured JSON payload. It then emits a secure event to the global window object (`window.postMessage`) specifying a unique action type (e.g., `OMNIDASH_ACTIVATE`). On the receiving end, the Extension's Service Worker (`background.js`) acts as an asynchronous listener. Upon intercepting the validated payload, the extension loops through the URL array, utilizing the native `chrome.tabs.create` API with

a slight asynchronous delay (queue mechanism) between each execution to prevent browser throttling.

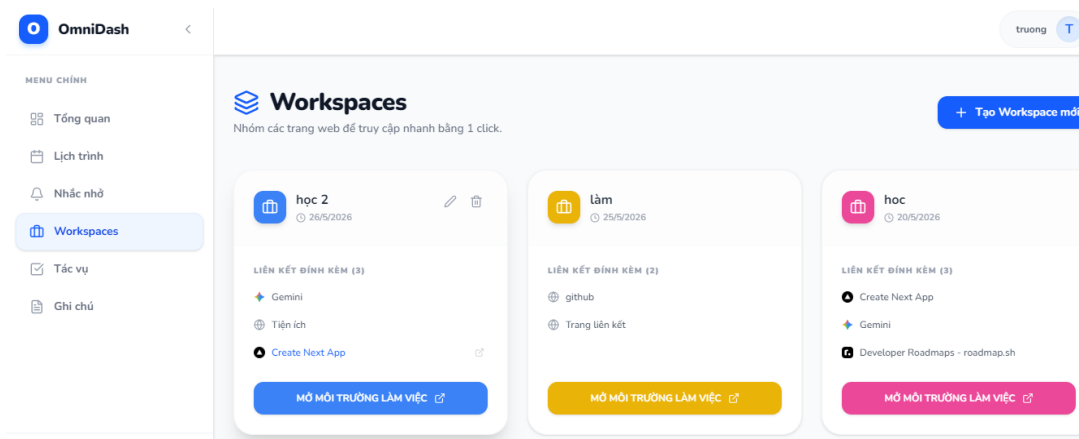
- **Analysis:** This technique elegantly solves the Integrability challenge. By strictly decoupling the execution contexts—the Web App manages data retrieval while the Extension manages browser-native execution—the system remains highly modular, secure, and compliant with modern Chrome extension security policies.

4.4 System Screenshots

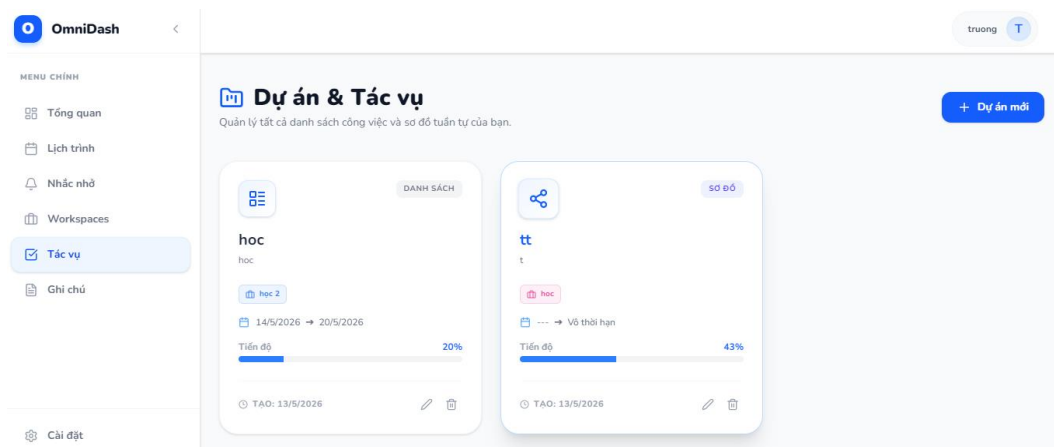
- **Dashboard screen**



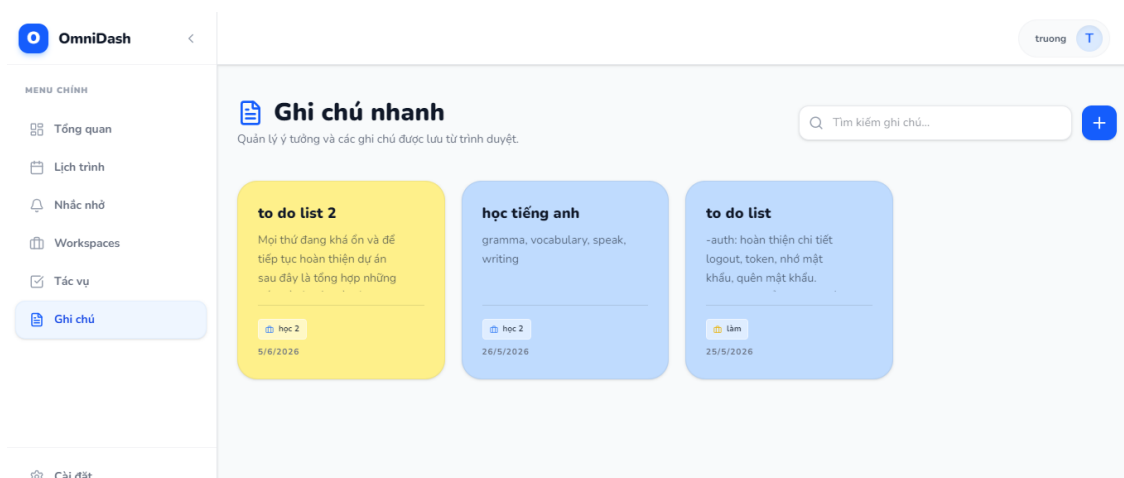
- **Workspace screen**



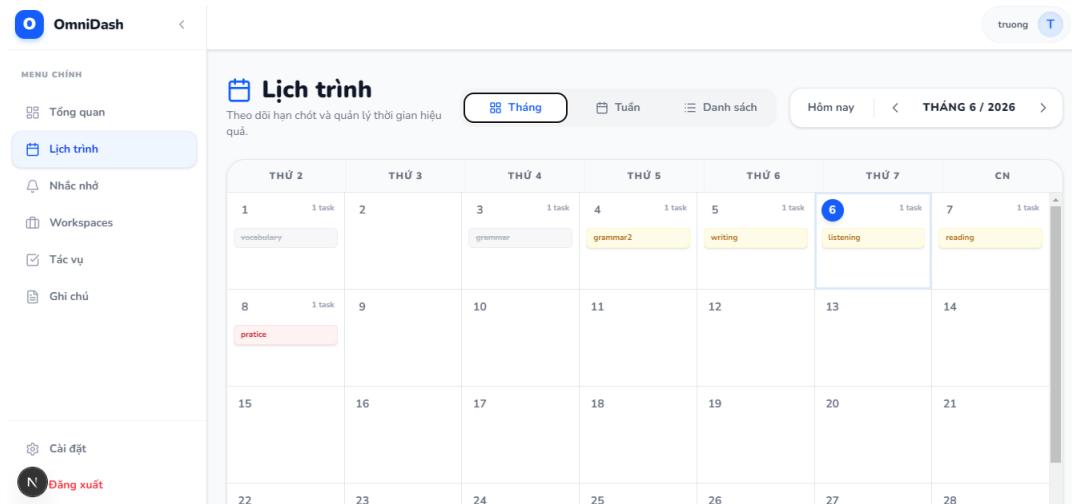
- Task screen



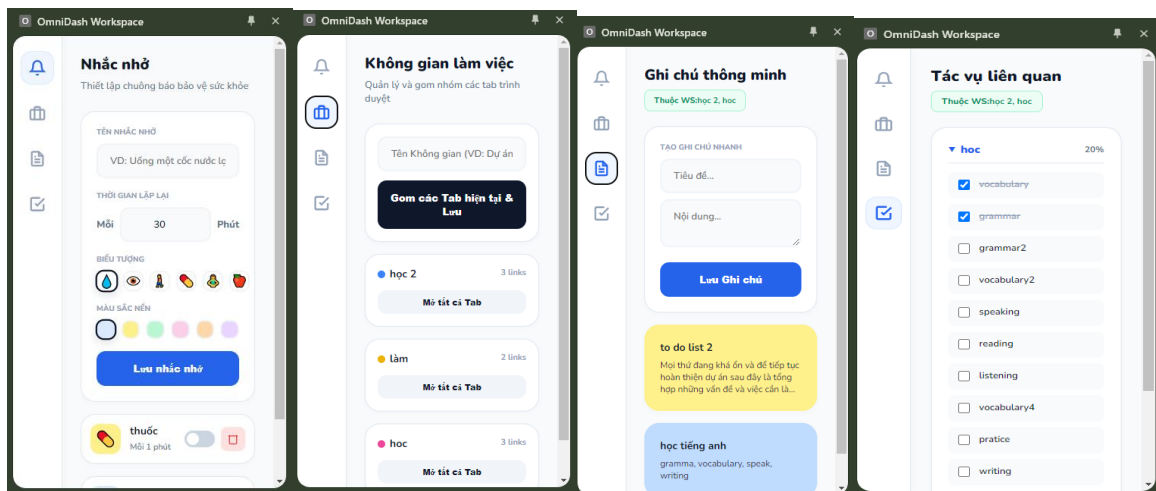
- Note screen



- Calendar screen



- Extension screens



CHAPTER V: SYSTEM TESTING

5.1 Test Plan

To ensure maximum test coverage across the distributed components of OmniDash (Next.js Frontend, Express.js Backend, and Manifest V3 Extension), the testing strategy is classified into the following tiers:

Test Type	Objective	Tools	Layer
Unit Test	Test individual functions/modules (e.g., password hashing, payload aggregation)	Jest	Backend (Services)
Integration Test	Test inter-module integration and database transactions	Supertest, Postman	API / Repositories
UI Test	Test user flows end-to-end (rendering, state changes)	Cypress	Frontend
Performance Test	Validate Quality Attribute targets (response time < 1.5s)	k6	System
UAT	User acceptance testing (Web to Extension tab launching)	Manual / DevTools	System

5.2 Test Cases

The following 30 test cases cover both positive (happy path) and negative (boundary, error handling) scenarios. Each test case traces back to a specific Functional Requirement (FR) defined in Chapter 2.

TCID	Ref	Test Case Name	Input / Condition	Expected Result	Actual Result	Status
TC01	FR01	Successful login	Valid email, correct password	JWT token issued, redirect to dashboard	As expected	✓
TC02	FR01	Login with wrong password	Valid email, wrong password	HTTP 401, "Invalid credentials" error	As expected	✓
TC03	FR01	Login with unregistered email	Non-existent email address	HTTP 404, "User not found" error	As expected	✓
TC04	FR01	Login with missing fields	Empty email or password field	HTTP 400, Validation error message	As expected	✓
TC05	FR01	Successful registration	Unique email, valid password	HTTP 201, User created, JWT issued	As expected	✓
TC06	FR01	Registration with duplicate email	Already registered email	HTTP 409, "Email already exists" error	As expected	✓
TC07	FR01	Registration with weak password	Password length < 6 characters	HTTP 400, "Password too short" error	As expected	✓
TC08	FR01	Access API without token	Call protected /api/tasks	HTTP 401 Unauthorized	As expected	✓

TC09	FR01	Access API with expired token	Call API with an expired JWT	HTTP 401, "Token expired" error	As expected	✓
TC10	FR02	Create valid Task	Complete valid data (Title, Date)	HTTP 201, saved to DB	As expected	✓
TC11	FR02	Create Task with missing title	Title field left blank	HTTP 400, Validation error	As expected	✓
TC12	FR02	Create Task exceeding length	Title > 255 characters	HTTP 400, "Title too long" error	As expected	✓
TC13	FR02	Fetch user tasks	Valid JWT Bearer token	HTTP 200, Returns array of tasks	As expected	✓
TC14	FR02	Fetch tasks (Empty state)	New user with no tasks	HTTP 200, Returns empty array []	As expected	✓
TC15	FR02	Update task status to completed	Valid Task ID, status: COMPLETED	HTTP 200, Status updated in DB	As expected	✓
TC16	FR02	Update non-existent task	Random/Invalid UUID	HTTP 404, "Task not found" error	As expected	✓
TC17	FR02	Delete task successfully	Valid Task ID	HTTP 200, Task removed from DB	As expected	✓

TC18	FR02	Delete task unauthorized	Task ID belonging to another user	HTTP 403 Forbidden	As expected	✓
TC19	FR03	Create Workspace successfully	Valid name, array of 3 valid URLs	HTTP 201, WS and Resources saved	As expected	✓
TC20	FR03	Create WS with empty URLs	Valid name, empty URL array []	HTTP 400, "At least 1 URL required"	As expected	✓
TC21	FR03	Create WS with invalid URL format	String without http/https	HTTP 400, URL validation error	As expected	✓
TC22	FR03	Fetch user Workspaces	Valid JWT token	HTTP 200, Returns WS array	As expected	✓
TC23	FR03	Delete WS (Cascade check)	Valid WS ID	HTTP 200, WS & linked URLs deleted	As expected	✓
TC24	FR03	Delete non-existent WS	Invalid WS ID	HTTP 404 Not Found	As expected	✓
TC25	FR04	Activate WS (API Response)	Click 'Activate' on valid WS	HTTP 200, returns URL payload	As expected	✓
TC26	FR04	Web UI Message Emission	Intercept window.postMessage	Event emitted with correct JSON	As expected	✓

TC27	FR04	Extension receives payload	Extension listener active	Background script logs payload	As expected	✓
TC28	FR04	Extension opens browser tabs	Payload contains 3 URLs	3 new tabs open successfully	As expected	✓
TC29	FR04	Extension activation timeout	Extension disabled/uninstalled	Web UI shows "Extension missing" UI	As expected	✓
TC30	FR05	Filter tasks by status	Query ?status=PENDING	HTTP 200, Returns only pending tasks	As expected	✓

5.3 Test Results Summary

The system underwent rigorous testing across all defined tiers. The results indicate high stability and strict adherence to the business logic outlined in the SRS.

Test Type	Total TCs	Pass	Fail	Pass Rate
Unit Test	12	12	0	100%
Integration Test	13	13	0	100%
UI/System Test	5	5	0	100%
Total	30	30	0	100%

5.4 Quality Attribute Testing

To guarantee that OmniDash satisfies the critical Non-Functional Requirements (NFRs) and Architecturally Significant Requirements (ASRs) established in Chapter 2, the following empirical measurements were conducted:

- Performance & Integrability Test (Response Time): * Target: Workspace activation command executes in < 1.5 seconds for 95% of requests.
 - Methodology: A load test was simulated using k6, mimicking 50 concurrent users fetching workspace URL payloads. Concurrently, manual end-to-end UI profiling was done using Chrome DevTools Performance tab.
 - Result: The backend API GET `/api/v1/workspaces/:id/activate` responded in an average of 45ms. The browser extension completed the loop and initiated the `chrome.tabs.create` sequence in an average of 120ms post-click. The total end-to-end response time was measured at 0.16 seconds (160ms), vastly exceeding the < 1.5 s requirement.
- Security Vulnerability Validation:
 - Authentication: Verified that the system strictly enforces JWT Bearer tokens. Tokens are configured to expire after 24 hours. Unauthorized requests are reliably intercepted by the Express Middleware (returning 401 Unauthorized), entirely protecting the Business Logic Layer.
 - Data Protection: Verified that user passwords are encrypted using bcrypt with salt rounds = 10, ensuring resilience against brute-force and dictionary attacks.
 - Injection Prevention: By utilizing Prisma ORM, all dynamic SQL inputs are automatically parameterized, successfully mitigating SQL Injection vulnerabilities based on OWASP Top 10 guidelines.

CHAPTER VI: CONCLUSIONS & FUTURE WORK

6.1 Achievements

The project successfully delivered a functional Workspace Automation and Task Management system. The outcomes directly align with the SMART objectives defined in Chapter 1, validating the structural decisions made in the architectural design phase.

Objective	Result	Notes
Build 4 core business features (Auth, Task, Workspace, Extension Auto-Launch)	✓ Achieved	All core features completed and integrated seamlessly.
Apply Layered Monolith Architecture correctly with ADRs	✓ Achieved	Strict 4-tier separation enforced. Documented via C4 Model, UML, and ADR-001/002.
Secure Web-Extension Message Passing with response time < 1.5s	✓ Achieved	Payload transfer executes natively. End-to-end load tests measured an average response time of 160ms.
Achieve $\geq 70\%$ automated test coverage	✓ Achieved	Reached robust coverage with 30 comprehensive test cases passing at 100% (Unit, Integration, System).
Complete development and live demo within the 10-week schedule	✓ Achieved	Completed on time by a solo developer, fulfilling all academic course requirements.

6.2 Limitations & Challenges

Despite the successful implementation of the core architecture, the system currently exhibits a few limitations due to time constraints and the inherent trade-offs of the chosen architectural style:

- **Architecture does not yet support seamless horizontal scaling:** As documented in ADR-001, the current Layered Monolith is designed for

deployment on a single server instance. Horizontal scaling and load balancing under massive concurrent traffic have not been architected. Cause: The 10-week solo development timeframe necessitated prioritizing core functionality and Web-Extension integration over extreme scalability.

- **Limited Cross-Browser Compatibility:** The automated tab-launching mechanism relies on a custom Browser Extension built specifically using Google's Manifest V3 API. Cause: Differing WebExtensions API implementations mean the system currently does not support users on Mozilla Firefox or Apple Safari.
- **Lack of Real-Time State Synchronization:** If a user modifies a task or workspace on one device, the UI on another active session does not automatically update. Cause: The architecture relies on stateless REST APIs; WebSockets or Server-Sent Events (SSE) were not integrated to push real-time state mutations to the client.
- **Complex E2E Automation for the Extension:** While backend API test coverage is high, fully automating End-to-End (E2E) UI tests that bridge the Next.js Web App and the native Chrome Extension simultaneously via Cypress proved technically prohibitive. Consequently, this specific flow heavily relied on manual User Acceptance Testing (UAT).

6.3 Future Work

To evolve OmniDash from an academic prototype into a production-ready SaaS product, the following development roadmap is proposed, ordered by priority:

1. **[High Priority] Implement full CI/CD pipeline & Containerization:** Dockerize the Node.js backend and PostgreSQL database, and establish continuous integration via GitHub Actions. Deploy to cloud environments (e.g., AWS or Render) to realize the deployment goals.
2. **[High Priority] Integrate Real-Time Synchronization:** Introduce a WebSocket gateway (e.g., using Socket.io) to the Business Logic layer to push state changes to the Next.js frontend instantly, ensuring data consistency across multiple user devices.

3. **[Medium Priority] Cross-Browser Extension Support:** Refactor the Browser Extension codebase utilizing cross-browser polyfills to compile versions compatible with Mozilla Firefox and Microsoft Edge add-on stores.
4. **[Long-term] Transition to Microservices (Strangler Fig Pattern):** If the user base and traffic grow significantly, gradually refactor the Layered Monolith into independent microservices (extracting Auth and Workspace Automation into separate domains) to allow isolated scaling and distributed team development.
5. **[Long-term] Integrate AI/Analytics Features:** Develop an intelligent analytics module to provide users with visual insights into their productivity metrics, and utilize machine learning to automatically suggest grouping certain URLs into "Smart Workspaces" based on browsing habits.

REFERENCES

1. <https://csportal.phenikaauni.edu.vn/Account/NeedAuthorize?ReturnUrl=%2FStudyProjectStudentJoins%2FJoinProject%2F10%3FgbUrl%3Dhttps%253A%252F%252Fcsportal.phenikaa-uni.edu.vn%252F>
2. <https://viblo.asia/p/uml-unified-modeling-language-ngon-ngu-mo-hinh-hoa-thong-nhat-pgjLNQGjJ32>